

Teleprompter Script: CS-PaperSum

Detailed presentation with integrated definitions

Welcome, everyone. Today we are going to dive deep into the scientific article CS-PaperSum, a revolutionary study that addresses the challenges of the exponential growth of literature in Computer Science.

To begin with, the central problem pointed out by the authors is that the expansion of academic research has made it unfeasible to track trends or extract insights manually. Traditional datasets provide only raw metadata, lacking structured summaries about the real contributions of the works.

The presented solution is CS-PaperSum, a large-scale dataset that gathers 91,919 articles extracted from 31 elite conferences, all enriched with structured summaries generated via ChatGPT. To validate the reliability of these summaries, the researchers used advanced artificial intelligence methodologies.

Among them, the use of **SciBERT** (which is a version of Google's BERT model pre-trained specifically on millions of scientific texts to accurately understand academic jargon) stands out. To analyze whether the generated summaries maintained the same essence as the original articles, they applied **t-SNE** (which is a statistical dimensionality reduction technique used to project complex data onto a 2D graph where similar elements are placed close together).

Furthermore, to ensure that the main terminologies were preserved, the team employed **KeyBERT** (an extraction tool based on machine learning that uses

embeddings to identify which words best represent the central concept of the text). In terms of data engineering, the automated extraction of the conclusion sections from the PDFs was performed with the help of **GROBID** (an open-source tool based on machine learning designed to convert academic documents into clean XML structures).

Moving into the results and case studies section, the paper maps the temporal evolution of major conferences in the field. At NeurIPS, for example, the focus shifted to graph neural networks and self-supervised learning. At AAAI, the priority became trustworthy AI and human-AI collaboration. At EMNLP, there was a drastic transformation towards large language models and the use of **RAG** (an AI technique that combines language models with external search systems to retrieve relevant documents in a database and generate more grounded responses). At ICCV, the highlight was the rise of vision transformers and **NeRFs** (which are neural radiance fields capable of reconstructing photorealistic three-dimensional scenes from two-dimensional images).

In conclusion, CS-PaperSum bridges the gap between raw bibliographic data and practical insights, opening doors for new summarization benchmarks and intelligent systems to support scientific discovery.

Thank you very much for following this reading. See you next time!